

Machine interpretation of facial expressions

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Facial expressions are an important source of information for human interaction. Therefore, it would be desirable if computers were able to use this information to interact more naturally with the user. However, facial expressions are not always unambiguously interpreted even by competent humans. Consequently, soft computing techniques in which interpretations are given some belief value would seem appropriate. This paper describes how the mass assignment approach to constructing fuzzy sets from probability distributions has been applied to the low-level classification of pixels into facial feature classes based on their colour. It will also describe how similar approaches can be used for the analysis of facial expressions themselves.

1. Introduction

Facial expressions are an important part of human/human interaction. Currently, however, the information which may be derived from these expressions is not used by computers for the following two reasons. Firstly, there is the formidable task of extracting a useful characterisation of a face from its image. Secondly, facial expressions and the emotional states or events that they signify are ambiguous and vague. This paper will discuss both of these aspects, along with how 'natural' facial features can be used in classifying and utilising expressions. A method for extracting facial features from an image which uses fuzzy sets derived from a small set of sample images will also be described.

1.1 The importance of facial expressions

When communicating with other humans we use many forms of communication. The most obvious is speech; however, this is often accompanied or supplanted by other methods. In particular, there is the tone with which someone says something as well as the facial and bodily expressions that accompany it. Indeed, when it comes to guessing how someone has reacted to something or how they are feeling, expression analysis is not only less intrusive, but also more accurate. If we consider the usual response to the question 'Are you OK?' (i.e. 'I'm fine'), we can see how little we use words to communicate our emotional state.

The information which we use all the time when communicating with others is currently not available to a computer. At present we interact with computers either via the keyboard or with the aid of a mouse. The interaction is one way and in the form of a command from the user to the computer. However, it would be more desirable if the computer could:

- initiate courses of action which are dependent on the mood of the user — in this case the facial expression would be used as a condition for undertaking the action,
- proactively suggest and implement commands which the user would approve of but has not yet specified — facial expression could be used here to monitor the user's response to the action.

This would extend the input modalities of the computer allowing a more 'natural' interaction with the user.

Further applications of an expression recogniser might also include a pain-detector in which the expression of a patient would be monitored in order to warn the carer, and a lip-reader to aid speech analysis.

1.2 The ambiguity of facial expressions

The human face can generate around 50 000 distinct facial expressions, which correspond to about 30 semantic distinctions [1]. Clearly the human face is extremely expressive. Given this, a number of outcomes are possible:

- we might expect a mapping of many expressions to a few emotions,
- we might expect that a given expression may be mapped to more than one emotional state, because the different states share the same expression,
- at any given time the subject may be in more than one emotional state, in which case we would expect a mapping from one expression to different emotional states.

Furthermore, different gradations of emotional states are experienced. For example we may be extremely angry or just a bit upset. We can therefore expect a degree of vagueness as to whether someone is in a particular emotional state or not.

2. Outline of a method for facial expression interpretation

Because of the ambiguity of facial expressions and the underlying emotions, interpretation of a given facial expression is not simply a matter of classification into one type. Rather we should expect it to belong to several classes with varying memberships. It is proposed here that a given expression should belong to one or more classes with a corresponding degree of support for each class. Thus an expression might be considered to be bored with a support of 0.8 and sad with a support of 0.4. A given expression can then be conveniently classified using rules, for example:

```
subject is happy
  if the lip shape is happy
    and the eye shape is happy
  with support S
```

The question then is how to assign supports for a given expression. This will depend on the measurable features selected from the face and the method of generating the support from these features.

2.1 Selecting facial features

The main muscles used in facial expressions are centred around the eye and the mouth. Consequently extracting measurements, such as width and height for these features will give a quantitative expression of their shape. Further information can be obtained from the eyebrows which reflect the muscle movement in the forehead more strongly than the eyes, but at this stage only the level of information that can be derived from the eyes and lips is being investigated.

2.2 Calculating the support

The information required for this application will come from various sources:

- visual information from the lips,
- visual information from the eyes,
- behavioural information about the user,
- other potential sources of information, e.g. keyboard typing speeds.

Using information from different sources can greatly increase the complexity of the classification since each extra source increases the dimensionality of the data.

Furthermore, for some of the sources data will be scarce. For example, behavioural information will come as and when the user performs certain series of actions. If, on the other hand, intermediate classifications for each source are used and then combined, less data will be required since classification is being done on a sub-space of the entire classification feature space and the complexity of each classification will be reduced. A Fril program will allow this to be done.

Intermediate classifications can be made using basic Fril rules in which the fuzzy sets have been generated from data according to mass assignment theory [2—5] using the data-browser [6, 7]. In this case the rules will have the form:

```
((shape of lip is lipshape1
  if width is shape1_lip_width
    height1 is shape1_lip_height1
    height2 is shape1_lip_height2
    height3 is shape1_lip_height3
    height4 is shape1_lip_height4))
```

where width, height1, etc, are the measured values from the image and shape1_lip_width, etc, are fuzzy sets generated by the data-browser. The resulting evidences can then be combined using either the evidential logic rule or the extended Fril rule.

The different sources may be combined in the form of a probabilistic decision tree. In this case the extended Fril rule would be used, which can be generated from data using the mass assignment ID3 algorithm [8]. An example of the rule might be:

```
((subject is happy given LIP_shape
EYE_shape)
  ((shape of LIP_shape is lipshape1)(shape
of EYE_shape is eyeshape1))
  ((shape of LIP_shape is lipshape1)(shape
of EYE_shape is eyeshape2))
  ((shape of LIP_shape is lipshape2)(shape
of EYE_shape is eyeshape1))
  ((shape of LIP_shape is lipshape2)(shape
of EYE_shape is eyeshape2))
):((0.8 0.9) (0.4 0.5)(0.7 0.9)(0.1 0.2))
```

Evidential logic rules, as described in Baldwin [9, 10] and, Baldwin et al [6], may be used if some sources are more discriminatory for a class than others. They can be generated from data using the data-browser and semantic discrimination [11]. In this case the rules would have the form:

```
((subject is happy given LIP_shape
EYE_shape)
  (most ( (shape of LIP_shape is
lip_shape1) 0.4
  (shape of EYE_shape is eye_shape1) 0.6)
  )
)
```

which can be read as: the subject is happy if it is mostly true that the lip shape is of type lip_shape1 and the eye shape is

of type `eye_shape1`, and where lip and eye shapes have weights of 0.4 and 0.6 respectively, i.e. the eye shape is more discriminatory for the class 'happy' than the lip shape.

3. Soft computing techniques for extraction of a facial feature from an image

3.1 The use of colour for facial feature extraction

The extraction of a face's facial features from an image is not a trivial task. As well as the problems encountered for any computer vision task, extracting facial features has the following problems:

- facial features are not geometrically simple,
- facial features vary in shape between individuals,
- facial features are highly deformable even for one individual,
- facial features should be extractable under varying lighting conditions.

As has been noted above, the features that need to be extracted are the eyes and mouth. Of these the mouth is particularly difficult to extract because the grey-scale values of the mouth in an image are typically very similar to those of the skin. Therefore, it is not possible to reliably find the boundaries of the lips from a grey-scale image. Consequently edge-detection on the grey-scale image is not a viable method.

However, these boundaries could be located if an appropriate transform could be found. In this case we are looking for a transform from the RGB values which would:

- vary according to the closeness of the pixel colour to the prototypical facial feature colour,
- be insensitive to changes in lighting conditions.

Similar approaches to feature extraction using chromaticity information can be found in Coianiz et al [12] and Yang et al [13].

3.2 Selecting an appropriate colour transformation

To reduce sensitivity to lighting conditions, an initial transform from RGB values was required to values in which the chromaticity is separated from the intensity. Many such transforms exist, such as HSV and YIQ values. However, it has been found that using normalised red and green values as described in Yang et al [13] has given the best results. In this case the initial transform is:

$$R_normalised = R/(R+G+B)$$

$$G_normalised = G/(R+G+B)$$

where R , G and B are the RGB values.

To meet the first condition examples of skin, lips and eyes were isolated by hand from a set of training images as

examples of facial features. The individual pixel values were converted into normalised red and normalised green, hue or IQ values. For each colour transform, fuzzy sets were formed for each facial feature using the data-browser method. The resulting image of the skin class can be seen in Figs 1—3. Here the intensity of the image corresponds to membership of the skin class so that a white pixel has high membership and a black pixel low membership. Clearly the normalised colour sets are more successful at separating skin from not-skin pixels, and this is, in general, the case for all classes. The normalised colour sets generated for the eye, lip and skin classes can be seen in Figs 4 and 5 where the y -axis corresponds to the membership of the set and the x -axis to the normalised colour.



Fig 1 Fuzzy image generated from hue values.



Fig 2 Fuzzy image generated from IQ values.



Fig 3 Fuzzy image generated from normalised colour values.

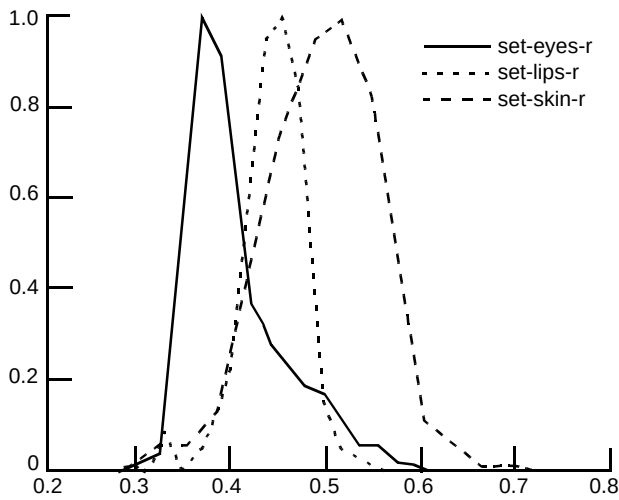


Fig 4 Fuzzy sets for normalised red.

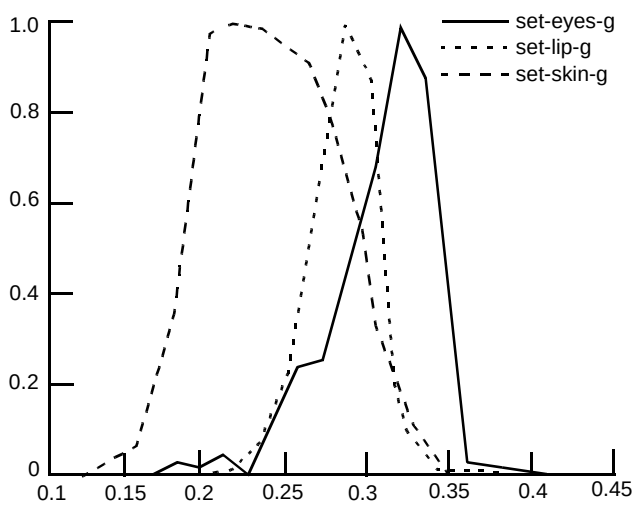


Fig 5 Fuzzy sets for normalised green.

Once these sets have been formed, support can be given for membership of a particular facial feature set for any novel pixel.

3.3 Resulting colour spaces

For illustrative purposes colour spaces for each class can be generated as shown in Figs 6—8. The axes are the normalised red and green values and the intensity is the rescaled product of the degree of membership in the normalised red and green fuzzy sets in such a way that white indicates the highest membership in the class and black indicates the lowest.

The most striking feature of these results is the surprising uniqueness of skin colour. It might be expected that the variation in skin colour would be greater than that of lip colour, but as can be seen in Fig 6, it occupies a very small area of the colour space in comparison to the lip class (Fig 7). It can also be seen that the eye space (Fig 8) is small and centred around the values of 0.333 for both normalised colours. This will correspond to a whitish colour

which is what should be expected. The smallness of the skin and eye spaces implies that both are good candidates for discrimination of classes on the basis of colour.

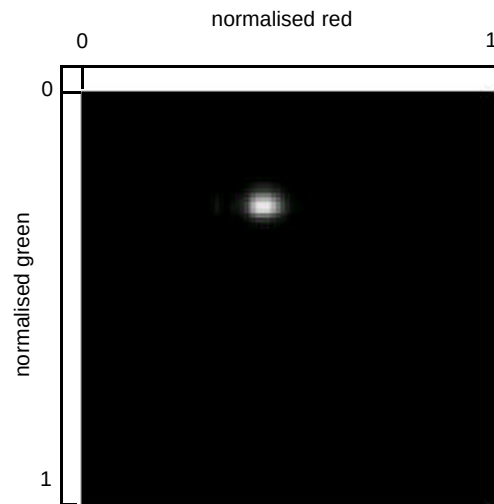


Fig 6 Colour space for skin.

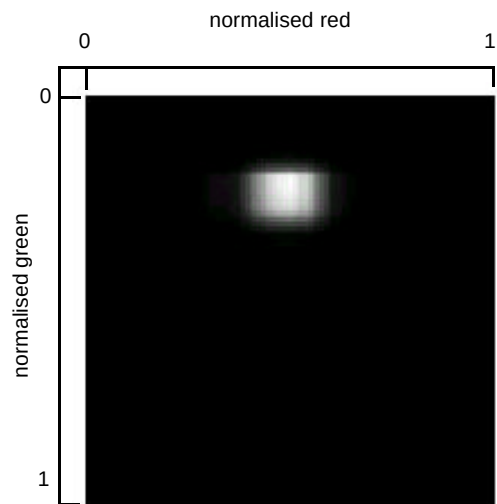


Fig 7 Colour space for lip.

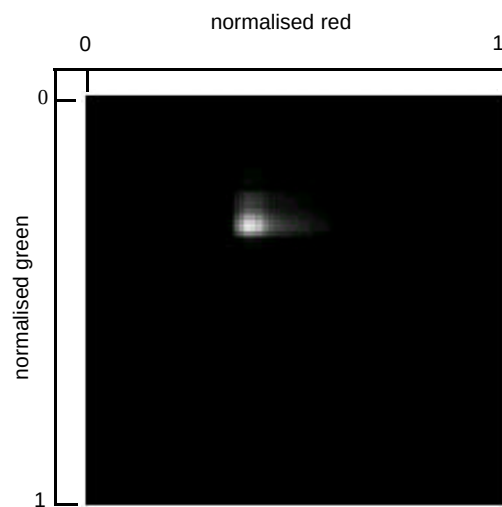


Fig 8 Colour space for eye.

3.4 Resulting fuzzy images

Figures 9—12 show the results of classifying pixels using the fuzzy sets. The degree of membership in each class is shown by the intensity of each pixel. As before, white pixels have a high membership and dark pixels have a low one.



Fig 9 Original image.



Fig 10 Skin image.



Fig 11 Lip image.



Fig 12 Eye image.

3.5 Improving the lip image

As can be seen from the resulting colour-maps and images there is a degree of intersection between the lip and skin pixel sets. An improvement to the lip image can be made by asserting that a pixel is a lip pixel if it belongs to the lip class and not to the skin class, i.e. lip pixel membership becomes:

$$(\chi_{\text{red-norm-lip}} * \chi_{\text{green-norm-lip}}) * (1 - (\chi_{\text{red-norm-skin}} * \chi_{\text{green-norm-skin}}))$$

where $\chi_{\text{red-norm-lip}}$ represents membership of the fuzzy set for normalised red for the lip class. The resulting image for the example can be seen in Fig 13.



Fig 13 Lip image after improvement.

3.6 Using intensity information

The use of intensity (defined as $0.3*R + 0.59*G + 0.11*B$) has been avoided so far because of its sensitivity to lighting conditions. This can be seen in Fig 14.

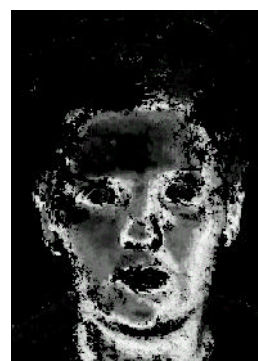


Fig 14 Image using intensity information which is sensitive to lighting conditions.

In Fig 14, the light source is on one side which has caused pixels on the left-hand side to have a lower membership in the skin class, but this lower membership is in fact due to differing lighting conditions across the face.

This illustrates the general point that using the data-browser method to generate fuzzy sets for intensity values is inappropriate because of the difficulty in obtaining enough data. Indeed, the training data would need to be evenly spread around the possible intensity space.

On the other hand, extreme intensity values would not be expected to belong to any of the classes since facial features are neither very reflective nor very absorbent. From small amounts of data, higher and lower bounds can be estimated for the intensity values. These can be implemented as crisp sets and used in addition to the normalised colour sets in classifying novel pixels.

Figures 15 and 16 show the improvement which can be made when using intensity information. Figure 12 shows the skin image generated only from the normalized colour values, and Fig 13 shows the same image generated from colour values with upper and lower bounds for intensity. In particular, removing extreme intensity values prevented misclassification of pixels which come from the hair in the original image.



Fig 15 Fuzzy image generated from normalised colour sets.



Fig 16 Fuzzy image generated from normalised colour sets and crisp intensity set.

3.7 Using positional information

As well as colour information, it is also known that the facial features all occupy regular positions relative to each other. Since skin pixels are so well classified, the expected positions of the other features relative to the centre of the skin can be used to improve their classifications.

The direction and the distance of objects, either from a certain point or from each other, must be known. For this, the centre of the skin image as defined by the average skin pixel co-ordinate is used as the centre point of the face. Since the distributions of the skin pixels in the x and y axes is expected to be symmetrical about the mean, the standard deviation is found from these axes, which is then used as a

scaling factor. Fuzzy sets are generated for the scaled distance of the lip pixels from the centre of face in x and y axes. These sets can be seen in Fig 17. The dashed lines represent the positional fuzzy sets for the eyes and the lighter lines represent those for the lips. The y axis corresponds to the membership of the class and the x axis corresponds to the rescaled distance. Again because of the lack of data the sets have been estimated.

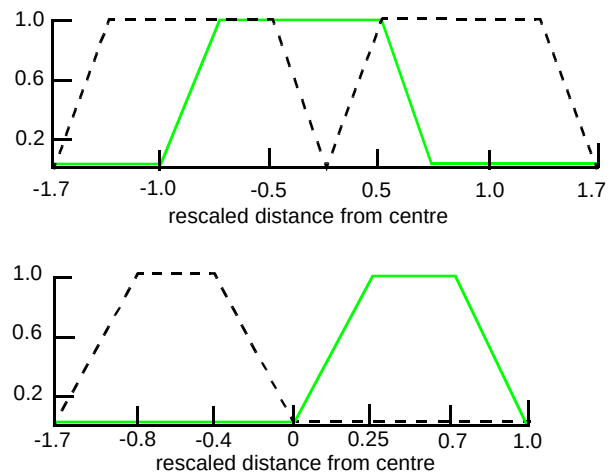


Fig 17 Fuzzy sets.

These sets are then used in conjunction with the normalised colour fuzzy sets and the crisp intensity sets in novel images. Figures 18 and 19 show the results of applying these sets to the sample image (see Fig 9).

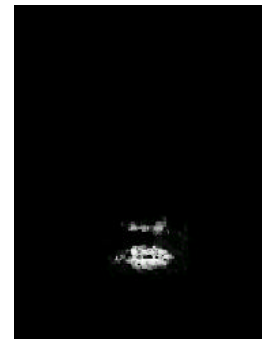


Fig 18 Lip image after combining the fuzzy set with the normalised colour and crisp intensity sets.



Fig 19 Eye image after combining the fuzzy set with the normalised colour and crisp intensity sets.

3.8 Removing final noise

As can be seen from Figs 18 and 19, the final fuzzy images may retain small mis-classified regions. However, it is known that the lips should form a solid object in the image plane. Therefore, the horizontal and vertical projections of an ideally classified image should cross the x or y value axis at only two points, i.e. non-zero x or y projection values should be consecutive. This can be seen in Figs 20 and 21, in which the abscissa corresponds to the vertical or horizontal projection value and the ordinate represents the x or y value in the image. For the purposes of this paper, consecutive non-zero projection values will be referred to as projection sections. In Figs 20 and 21 there is one section per image starting at C_1 and ending at C_2 .

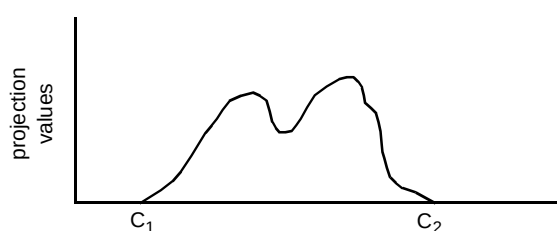


Fig 20 y value in image.

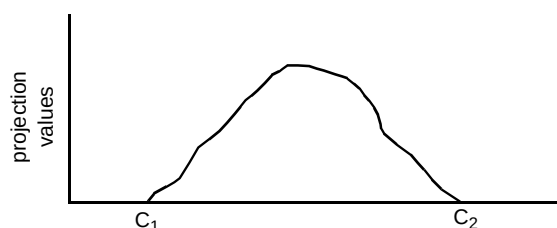


Fig 21 x value in image.

Consequently, if there is more than one section in either projection, one of these sections can be considered to be misclassified. The smaller section is assumed to be misclassified and any of the pixels that fall within this range are removed. This is repeated iteratively on the x and y axes until no more pixels are removed. This will remove any misclassified region which is separated from the correctly classified region in either the x or y directions.

3.9 Finding the final shape

It is now necessary to select an appropriate representation for the isolated lip pixels. This representation should be expressive enough to capture the important mouth movements and simple enough to allow easy classification of these movements. A parameterised model of the mouth, similar to those used in Coianiz et al [12] and Rao and Mersereau [14], has been chosen.

The final lip shape is defined as four half ellipses which meet at the corners of the mouth, as illustrated in Fig 22. Solid lines show the actual lip shape. Dotted lines show the model. The shape has five parameters — the height of each ellipse and the width which is the same for all the ellipses.

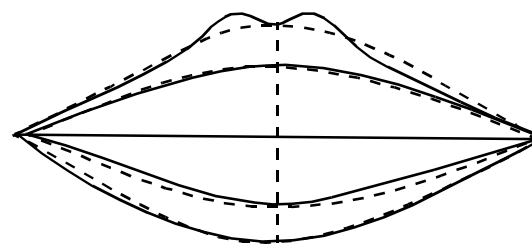


Fig 22 Lip shape.

This should express the height of each lip, the separation of the lips and the width of the whole mouth.

To extract these parameters firstly the corners of the lips are found. This is achieved by finding the zero crossings of the horizontal projection of the final fuzzy lip image (C_1 and C_2 in Fig 21), and then finding the average y value of the lip pixels with these x values. The lip area is then re-oriented so that the lip corners are level.

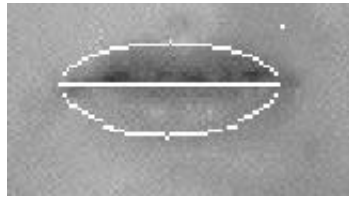
Once the corners have been correctly oriented, the point at which the centre of each lip starts and finishes can be found. A five-pixel wide vertical strip is made through the middle of the lips. Zero-crossings of this strip are then considered to be potential height parameter values. To decide which combination of parameters is correct a score is assigned to each one. Since the parameterised model should show where lip and not-lip pixels should be, the score is simply the sum of those pixels within the lip area in the parameterised model which are also classified as lip in the fuzzy image and those pixels which are outside or between the lips in the parameterised model which are also classified as not-lip in the fuzzy image. The best scoring combination of parameters is considered to be the final lip shape.

3.10 Results

The method was tested on a set of images of subjects taken under uncontrolled indoor lighting conditions with a flash camera. These mug-shots were passed on to the lip extraction system as whole images. The test images and their results can be seen in Figs 23—26. The white lines in the images are where the lip extraction system considers the lips to be, and close-ups of the mouth area are shown alongside the test images.



Fig 23



Sample image 1.

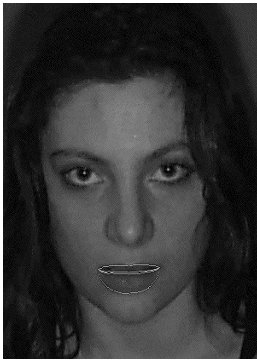
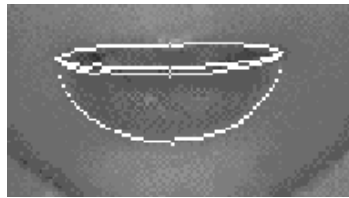


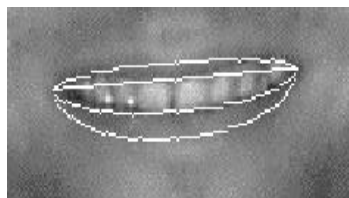
Fig 24



Sample image 2.



Fig 25



Sample image 3.



Fig 26



Sample image 4.

The sample images show both open (Figs 23, 25 and 26) and closed lips (Fig 24) as well as lip areas that are not upright in the xy plane (Fig 26). In general, the results show a good fit to the image, particularly the outer lip edges.

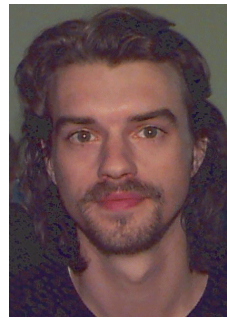
4. Conclusions

This paper has outlined the ambiguous nature of facial expressions and drawn attention to the possible causes of these ambiguities. A set of methods has been proposed with which expressions can be characterised and which reflects this indeterminacy. At present, no results are available for assessing these methods because of the difficult problem of extracting the necessary information from an image. However, a method has been developed for extracting an expressive and useful characterisation of facial features from an image which requires a very small set of training images. This method has been applied to the extraction of the lip shape from an image. In practice, it has been shown to be reliable, accurate and fairly robust. Further work will now concentrate on applying this method to the extraction of the eye shapes and using these extracted feature shapes along with the proposed methods to categorise and use the facial expressions.

References

- 1 Terzopoulos D and Waters K: 'Analysis and synthesis of facial image sequences using physical and anatomical models', IEEE, PAMI, 15, No 6 (June 1993).
- 2 Baldwin J F: 'Combining evidences for evidential reasoning', International J of Intelligent Systems, 6, pp 569—616 (1991).
- 3 Baldwin J F: 'A theory of mass assignments for artificial intelligence', Proc IJCAI Workshop on Fuzzy Logic, Australia (1991).
- 4 Baldwin J F: 'A new approach to inference under uncertainty for knowledge-based systems', in Krues R and Siegel P (Eds): 'Symbolic and quantative approaches to uncertainty', Springer Verlag Lecture Notes in Computer Science, No 548, pp 107—115 (1991).
- 5 Baldwin J F: 'The management of fuzzy and probabilistic uncertainties for knowledge-based systems', in Shapiro S (Ed): 'Encyclopedia of AI', John Wiley (2nd ed), pp 528—337 (1992).
- 6 Baldwin J F, Martin T P and Pilsworth B W: 'Fril-Fuzzy and Evidential Reasoning in Artificial Intelligence', Research Studies Press (1995).
- 7 Baldwin J F and Martin T P: 'A fuzzy data-browser in FRIL', in Baldwin J F (Ed): 'Fuzzy Logic', John Wiley & Sons (1996).
- 8 Baldwin J F, Lawry J and Martin T P: 'A mass assignment based ID3 algorithm for decision tree induction', International J of Intelligent Systems, 12, No 7, pp 523—552 (1997).
- 9 Baldwin J F: 'Evidential support logic programming', Fuzzy Sets and Systems, 24, pp 1—26 (1987).

- 10 Baldwin J F: 'Evidential reasoning under probabilistic and fuzzy uncertainties', in Yager R R and Zadeh L A (Eds): 'An introduction to fuzzy logic and applications in intelligent systems', Kluwer Academic Publishers (1991).
- 11 Baldwin J F: 'Knowledge from Data Using Fril and Fuzzy Methods', in Baldwin J F (Ed): 'Fuzzy logic', John Wiley & Sons (1996).
- 12 Coianiz T, Torresani L and Caprile B: '2D deformable models for visual speech analysis', Technical Report, Instituto per la Ricerca Scientifica e Tecnologica (1996).
- 13 Yang J, Wu L and Waibel A: 'Focus of attention in video conferencing', Technical Report, CMU-CS-96-150, School of Computer Science, Carnegie Mellon University (June 1996).
- 14 Rao B D and Mersereau R M: 'Lip modelling for visual speech recognition', Technical Report, School of Electrical Engineering, Georgia Institute of Technology (1996).



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